



A THOUGHT EXPERIMENT IN PRICE THEORY

The Real and the Imaginary

On pricing bitcoin as a complex number, and what gets lost when a market collapses two axes into one

$$z = R + Ii$$

The Real and the Imaginary

On pricing bitcoin as a complex number

TL;DR

- Markets usually treat "fundamentals" and "speculation" as a single blended number — 30% real, 70% hype. That framing loses information, because the two don't actually blend; they sit at right angles to each other.
- Model the price of bitcoin as a complex number, $z = R + Ii$. R , the real part, is what would remain true with every trader removed: issuance schedule, hash rate, settlement finality, self-custody, Lightning capacity. I , the imaginary part, is everything reflexive: leverage, ETF flow momentum, the "digital gold" narrative, belief in belief.
- The quoted price is the **modulus**, $|z| = \sqrt{R^2 + I^2}$ — magnitude only. The market's mood is the **argument**, $\theta = \arctan(I/R)$ — how much of that magnitude is narrative versus network.
- Multiplying by i rotates a vector 90° without changing its length — a workable metaphor for a halving or an ETF approval: the size of the move is one question, what the move is made of is another.
- Bull and bear cases typically share the same R and disagree only on the sign of I — they are **complex conjugates** of each other, z and $\bar{z}$, and $z \cdot \bar{z}$ collapses back to $|z|^2$, the market cap squared, regardless of who is right about sentiment.
- The essay closes by taking the metaphor apart: θ cannot actually be computed, R is not as objective as it is made to sound, and multiplication by i describes no real market operation. This is a lens for organising disagreement, not a valuation model.

§ Introduction and Scope

Every argument about bitcoin's price eventually arrives at the same fork. One side says the number on the screen reflects something real — scarce issuance, a growing settlement network, a supply that cannot be voted on by a committee. The other side says the number is mostly narrative — leverage, momentum, a story that six different marketing departments are currently selling in six different accents. Both sides are usually right about their half, and both sides usually try to resolve the disagreement the same way: by blending the two into a single percentage. "Bitcoin is 40% fundamentals, 60% hype," someone will say, as though value were a cocktail that could be mixed to taste.

That blending move is the actual error, not either side's diagnosis. Fundamentals and narrative are not the same kind of quantity added together — they don't share a unit, and one does not erode the other. A speculative premium does not "dilute" a fundamental floor the way water dilutes wine; it sits on an entirely different axis, capable of swinging from euphoric to hostile in a week while the network's hash rate, issuance schedule and settlement finality do not move at all. Whatever description tries to hold both, needs somewhere to put a quantity that is real, distinguishable from the first, and irreducible to it.

That is precisely what the imaginary unit is for. This essay develops a single idea: treat the price of bitcoin as a complex number, $z = R + I i$, where R is the axis a forensic accountant could verify from the network alone, and I is the axis a behavioural economist would recognise as reflexive belief. The point is not that bitcoin's price literally has an imaginary component in any measurable sense — nobody is proposing a spreadsheet column labelled "i". The point is that the algebra of complex numbers already contains the right shape for this problem: two independent axes, a magnitude that hides their composition, and an angle that reveals it. Complex numbers are borrowed here the way "duration" is borrowed from calculus to price a bond — as a working lens, not a claim about ontology. Section 7 takes the lens apart and is honest about where it stops holding. Everything before that is the case for picking it up in the first place.

1 A Complex Number in Sixty Seconds

A complex number $z = a + b i$ has two independent components living on perpendicular axes: a real part a , and an imaginary part b , where i is defined by the property $i^2 = -1$ — a number that, when squared, points backwards along the real line rather than forwards, which is another way of saying it points somewhere the real line cannot reach on its own. Plotted on the Argand plane, a sits on the horizontal axis and b on the vertical one, and z is the point — or equivalently the vector — at coordinates (a, b) .

Two derived quantities matter for what follows. The **modulus**, $|z| = \sqrt{a^2 + b^2}$, is the vector's length: the single number you get by collapsing both axes into one, which is exactly what a market price does every time it is quoted as a dollar figure. The **argument**, $\theta = \arctan(b/a)$, is the angle the vector makes with the real axis: it says nothing about size and everything about composition — how much of that length runs along the real axis versus the imaginary one.

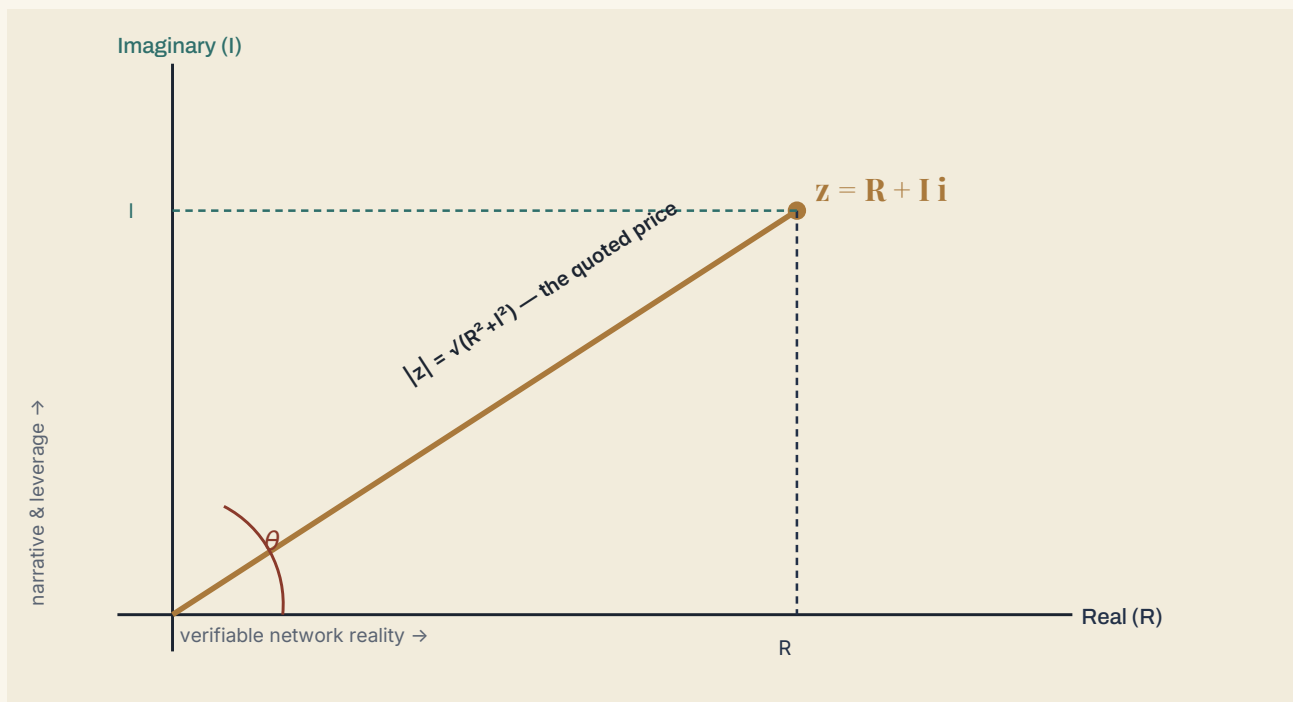


Figure 1. The Argand plane, relabelled. What the market calls "the price" is the length of z — the modulus. What the market actually disagrees about is θ , the angle — how much of that length is network reality and how much is narrative.

2 Two Axes for One Price

Assembling R is a discipline exercise: it should only contain what would still be verifiably true if every speculator vanished overnight and the order books emptied out. Assembling I is the opposite exercise: it should contain everything that depends on what other people currently believe, and would evaporate the instant that belief did.

AXIS	CONSTITUENTS
R — real part	Fixed 21m issuance schedule and halving-enforced disinflation; network hash rate and the sunk energy cost behind it; count of UTXOs held in long-term self-custody; Lightning Network routing capacity; settlement finality — the property that a confirmed transaction cannot be reversed by any authority.
I — imaginary part	Open interest and funding rates in the derivatives market; spot ETF creation/redemption flow momentum; the prevailing macro-liquidity narrative (rate-cut expectations, dollar strength); "digital gold" and reserve-asset storytelling; plain reflexivity — price rising because it rose, in the sense George Soros used the word.

Neither axis is static, and neither is pure. Hash rate is itself purchased with capital that has its own opportunity cost, which is a seed of the objection developed in Section 7. But the split earns its keep immediately: it explains why a piece of bearish news can hit a market that has changed nothing about its issuance schedule or its settlement guarantees, and why a chart can go essentially vertical without the network processing a single additional transaction. Those are movements almost entirely along I, with R barely touched — and a framework with only one axis has no vocabulary for that distinction. It can only report that "the price moved," which is the least interesting fact about any given move.

A speculative premium does not dilute a fundamental floor the way water dilutes wine. It sits on a different axis entirely — capable of swinging from euphoric to hostile in a week while the thing it is layered on top of does not move at all.

3 What the Modulus Hides

Every number that gets quoted — \$69,044, \$124,128 — is $|z|$: magnitude only, with R and I already folded together and the seam invisible. That collapse is not a flaw in how prices are quoted; a single scalar is what a market needs to clear a trade. But it does mean the interesting variable, θ , has to be reconstructed rather than read off, and reconstructing it is necessarily an interpretive act, not a measurement — there is no independent instrument that prices I in dollars the way a hash-rate query prices R in exahashes. What can be done honestly is to look at bitcoin's major cycle peaks and ask, qualitatively, how much of each one looks like it was R doing the work versus I doing the work.

The four peaks below are real dates and real prices. Their placement on the R-I plane is not a fitted or measured value — no such measurement exists — it is a schematic reading, offered as an argument to disagree with rather than a data series to trust.

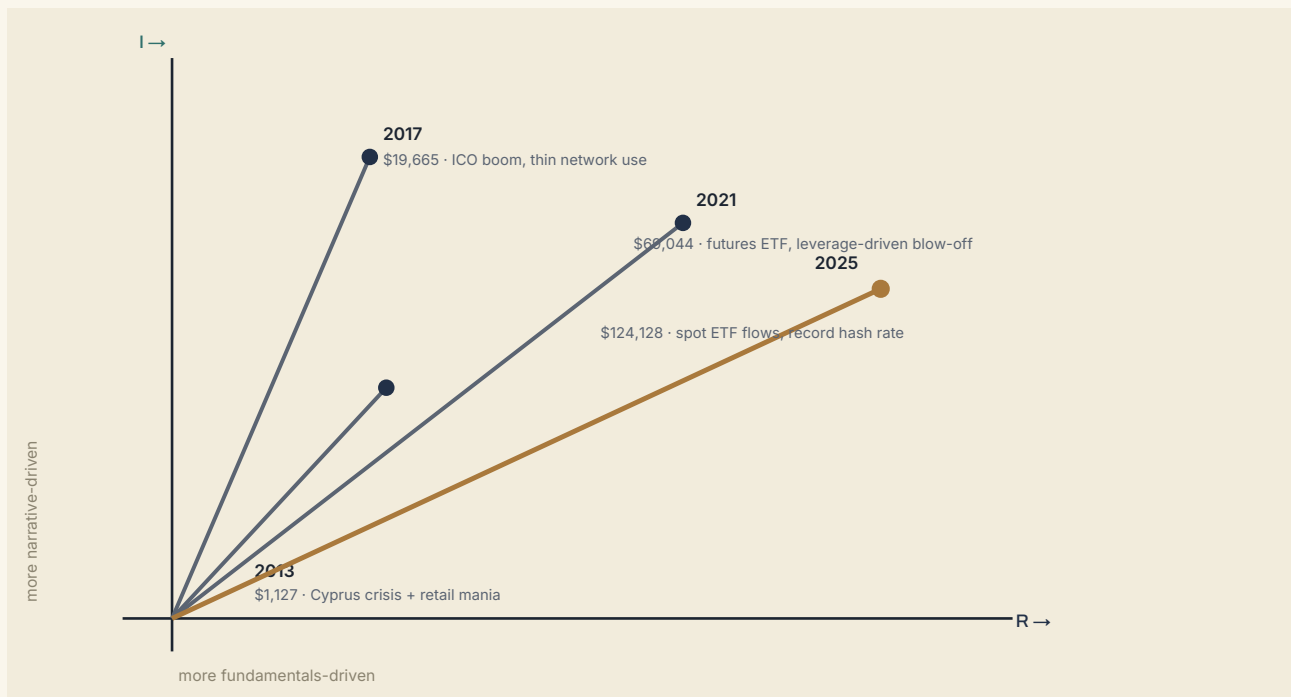


Figure 2. Schematic, not measured — illustrative positioning, not a computed θ . Dates and USD peak prices are historical (bitcoin's Nov 2013, Dec 2017, Nov 2021 and Aug 2025 cycle tops); their placement on the R-I plane is an interpretive argument, not a data series.

The reading the diagram makes is this: 2017 sits closest to the I axis because almost none of that peak was accompanied by growth in real network usage — it was retail mania riding an ICO wave with a token that had nothing to do with bitcoin's own settlement layer. 2021 drifts further along R because institutional custody infrastructure, futures markets and genuine treasury adoption existed by then, even though leverage still did a great deal of the work getting to \$69,044. 2025 sits furthest along R of the four, not because narrative was absent — spot ETF inflows are pure I, a bet on the story wrapped in a regulated wrapper — but because the network underneath it, measured in hash rate and settled value, had genuinely grown between cycles. None of this is a forecast or a model output. It is what the exercise of trying to plot θ forces you to notice, which is closer to the exercise's value than any resulting picture.

4 Rotation as Regime Change

Multiplying a complex number by i rotates it 90° counter-clockwise without changing its length: $i \cdot (a + bi) = -b + ai$. The magnitude $|z|$ is identical before and after; only the composition has changed. That is a strange and useful thing for an event to do to a market, and a small number of bitcoin's structural events plausibly resemble it more than they resemble ordinary appreciation.

The clearest candidate is the January 2024 approval of spot bitcoin ETFs in the United States. It did not alter issuance, hash rate or settlement finality on the day it happened — the real axis was untouched. What it did was change who could hold exposure and through what wrapper, converting a

chunk of what had been narrative-and-access-constrained demand into narrative-and-regulated demand. That is not obviously a move along R or a move along I in isolation; it is closer to a rotation — the same underlying asset, repriced through a different composition of belief about who is allowed to hold it. A halving is the harder case to place, and arguably the wrong candidate for a pure rotation: it changes the real issuance schedule directly, so it is at least partly a move along R, with any subsequent price action running mostly through I as the market narrativises the supply shock in advance and after the fact. Rotation is the more interesting metaphor precisely because it does not fit every event — it is worth reaching for only when magnitude looks unchanged but composition looks disturbed, which is a much narrower claim than "the price moved."

5 Fiat's Own z

The framework is not built to flatter bitcoin, and it embarrasses fiat currency in roughly the same shape. A fiat currency also has a real part and an imaginary part; it simply draws them from different sources. Its R is bounded by the productive capacity and tax base of the issuing state — the goods and services actually producible within its jurisdiction, the government's real ability to extract resources through taxation. Its I is trust in the central bank's independence, the sovereign's credibility, reserve-currency inertia, and geopolitical habit — the pure willingness of the rest of the world to keep pricing oil and settling trade in it out of custom rather than necessity.

Bitcoin's pitch, read through this lens, is not that its own I is zero — no liquid, freely traded asset has an imaginary part of zero, and a market with $\theta = 0$ would be a market nobody was pricing at all. The pitch is narrower and more defensible: that its R is unusually hard, auditable in real time by anyone running a node, and immune to unilateral revision by the single institution that happens to control it — where fiat's R is itself substantially a confidence trick, since a tax base is only as real as a population's willingness to keep paying and a government's willingness to keep collecting. Put the two side by side and the actual argument is about which currency's real axis is more trustworthy, not about which currency has escaped the imaginary axis altogether. Nobody has.

6 The Conjugate: Bull and Bear Are the Same Number

The complex conjugate of $z = R + Ii$ is $\bar{z} = R - Ii$ — identical real part, imaginary part flipped in sign. This is the cleanest part of the whole framework, because it describes something that actually happens in every serious argument about bitcoin's price: the bull and the bear are usually not disputing R at all. They are looking at the same hash rate, the same issuance schedule, the same self-custody figures, and assigning opposite signs to what those facts mean for the narrative axis.

WORKED EXAMPLE

A bull points to growing self-custody as evidence of conviction — a positive contribution to I : holders who won't sell regardless of price. A bear points to the identical growing self-custody figure as evidence of an illiquid, thinly-traded market vulnerable to a violent unwind on the first large sell order — a negative contribution to I . Same R . Same fact. Opposite sign on the imaginary part built on top of it.

Multiply the conjugate pair together and something clarifying happens: $z \cdot \bar{z} = (R + Ii)(R - Ii) = R^2 + I^2 = |z|^2$. The imaginary parts cancel entirely — the disagreement itself has no effect on the modulus squared, which is just the market capitalisation, squared, regardless of who wins the argument about sentiment. This is not a claim that bull and bear cases are equally correct, or that sentiment doesn't matter — I clearly moves $|z|$ enormously in the short run, which is the entire subject of Section 3. It is a narrower and more useful observation: that the shouting match over θ is very often two parties looking at exactly the same R and disagreeing only about which direction to point the arrow, which is a much smaller and more tractable disagreement than either side usually presents it as.

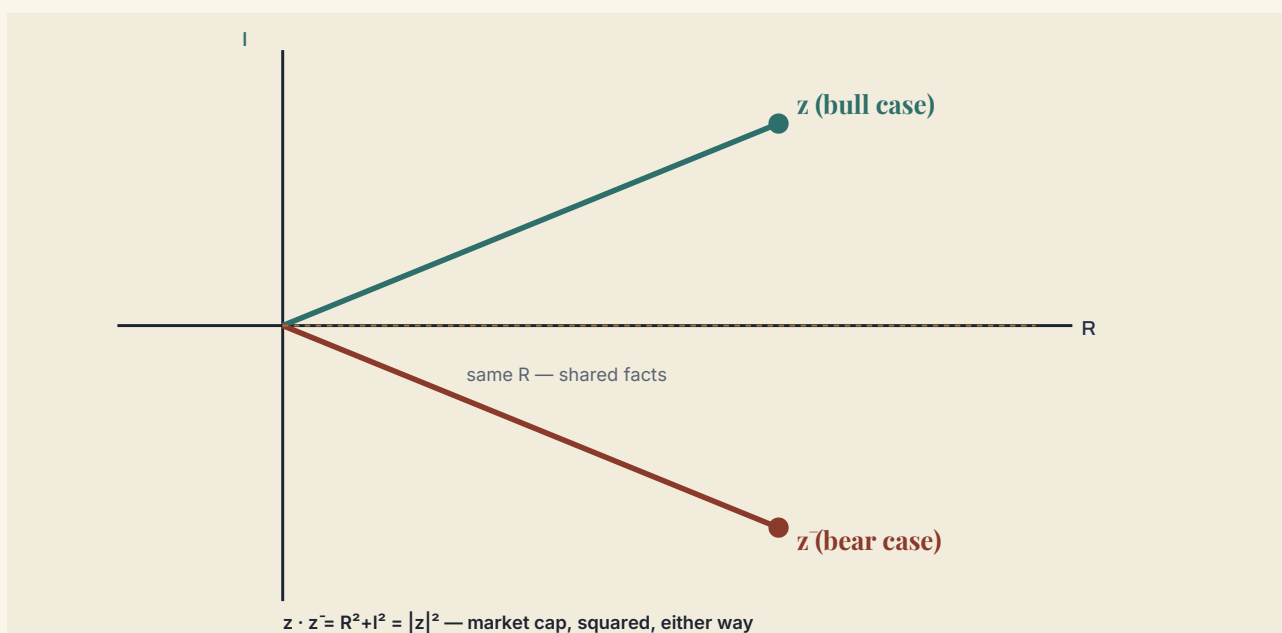


Figure 3. The bull case and the bear case as conjugate pairs — mirrored across the real axis, sharing every fact about R , disagreeing only about the sign of I .

7 Where the Metaphor Breaks

A framework this tidy should be distrusted a little, and the distrust is worth spelling out rather than leaving implicit.

LIMITS OF THE MODEL

- **θ is not actually computable.** There is no independent instrument that prices the imaginary axis in dollars separately from the real one — Section 3's diagram is an argument, not a measurement, and should never be mistaken for one. Anyone who claims to have computed a market's precise "sentiment angle" is manufacturing false precision.
- **R is not as objective as it sounds.** Hash rate is purchased with capital, and that capital has its own speculative motive — miners lever up against expected future prices, meaning the "real" axis has a narrative axis quietly embedded inside it. The split is a modelling choice, useful for organising an argument, not a fact discovered in the data.
- **Rotation is a metaphor, not a mechanism.** No market operation literally multiplies a price by i . Section 4's reading of the ETF approval is an interpretive claim about composition, offered because it is a more precise question than "did the price go up," not because complex multiplication describes anything an order book actually does.
- **The conjugate simplifies real disagreement.** Not every bull/bear dispute is actually a same-R, opposite-sign-I disagreement — some genuinely dispute the facts about R itself (whether self-custody figures are accurately measured, whether hash rate estimates are reliable). The conjugate model applies cleanly only to the subset of arguments that are actually about interpretation rather than fact.

None of this sinks the exercise; it bounds it. The value of treating price as $z = R + Ii$ was never that it produces a number nobody had before. It is that it forces every claim about "value" to declare which axis it is actually making a claim about, and stops the two axes from being silently averaged into a figure that answers neither question. That is a modest, worthwhile thing for a piece of financial rhetoric to do — considerably more modest than the framework's own presentation might suggest, which is itself a small demonstration of the difference between R and I.

§ Summary

Bitcoin's price is conventionally reported as a single number, which forces two genuinely different kinds of information — verifiable network reality and reflexive narrative — into one figure that cannot distinguish between them. Modelling price as a complex number, $z = R + Ii$, restores the distinction: R holds what would survive the disappearance of every speculator (issuance, hash rate, settlement, self-custody, Lightning capacity), and I holds what would not (leverage, flow momentum, story, belief in belief). The quoted price is the modulus, $|z|$; the market's mood is the argument, θ . Rotation by i offers a narrow, useful description of events like the 2024 spot ETF approval, where magnitude looked steady but composition plausibly shifted. Bull and bear cases are frequently conjugate pairs, z and $\bar{z}$, sharing R and disputing only the sign of I — and multiplying the pair together collapses the dispute back to $|z|^2$, the market cap, squared, regardless of who is right. The metaphor is deliberately taken apart in the essay's final section: θ cannot be measured, R is not as clean as it is made to sound, and no market mechanism literally performs complex multiplication. What the framework earns, modestly, is a discipline — it makes every argument about bitcoin's "true value" specify which axis it is actually arguing about, rather than averaging two incompatible quantities into a figure that settles nothing.

§ References

1. Nakamoto, S. (2008). Bitcoin: A Peer-to-Peer Electronic Cash System. — issuance schedule and settlement design underlying the R axis.
2. Soros, G. (1987/2003). *The Alchemy of Finance*. Wiley — the theory of reflexivity underlying the treatment of I as self-reinforcing belief rather than noise.
3. Cambridge Centre for Alternative Finance. Cambridge Bitcoin Electricity Consumption Index (CBECI) — ongoing hash rate and energy-cost estimation used as an R-axis reference.
4. CoinGecko Research (2025). When Do Bitcoin All-Time Highs Happen? — post-halving cycle-top timing and prices for 2013, 2017, 2021 and 2025 cited in Figure 2.
5. U.S. Securities and Exchange Commission (January 2024). Approval order, spot bitcoin exchange-traded products — the regime-change event discussed in Section 4.
6. Plan B (2019). *Modelling Bitcoin's Value with Scarcity*. — an early attempt to formalise a purely supply-side (R-axis) valuation model; cited here as a useful foil, not an endorsement — its predictive track record after 2021 is widely disputed and it is not relied upon for any figure in this essay.
7. Shiller, R. (2000). *Irrational Exuberance*. Princeton University Press — narrative-driven price formation as a general market phenomenon, of which the I axis is a bitcoin-specific instance.
8. Ratty2Tails (Monetary History Series). *The Stone That Cannot Be Quarried* — companion essay on bitcoin's fixed supply and price discovery mechanics; treats in scalar form much of what R denotes here.